Nicholas C. Taylor

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Research Interests

I am interested in spatial and spatiotemporal statistical modeling, predictive model validation, satellite remote sensing, and functional data analysis, with applications to water resources and the environmental and biological sciences. My work is motivated by applied problems, and I enjoy collaborative research that brings statistical methods to bear on real-world questions.

Education

Ph.D. in Statistics (GPA: 3.84) — Expected: 2027
Baylor University; Waco, TX
Department of Statistical Science
Advisor: Dr. Amanda S. Hering

M.S. in Statistics — May 2025
Baylor University; Waco, TX
Department of Statistical Science

M.S. in Mathematics (GPA: 3.72) — May 2023
Texas Tech University; Lubbock, TX
Department of Mathematics and Statistics

B.S. in Mathematics (GPA: 3.96) — May 2021
University of Mary Hardin-Baylor; Belton, TX
Department of Mathematics
Minor in Computer Science
Summa Cum Laude

Publications

Taylor, N. C., Macleod, M. S., Powers, S. M., & Hering, A. S. (2026). *Predicting in-lake turbidity using satellite data: Creating boat paths to compare cross-validation strategies* [In Preparation].

Weix, D., Taylor, N. C., Kim, R., Cath, T. Y., & Hering, A. S. (2026). *Simultaneous mean and sparse covariance monitoring via maximum likelihood estimation with positive-definite thresholding* [In Revision for *Quality Engineering*].

Teaching Experience

Instructor of Record (Graduate Student) — Aug 2024 - Dec 2025
Baylor University; Waco, TX
Fully responsible for introductory statistics courses, developing lectures, assessments, and holding regular office hours for approximately 30 students per term.
Fall 2024, Spring 2025, & Fall 2025 - STA 1380: Introduction to Statistics
- Department of Statistical Science Outstanding Teacher Award, 2024-2025
- Recognition for Excellence in Teaching, Spring 2025 & Fall 2025

Graduate Assistant - Supplemental Instructor & Teaching Assistant July 2023 - May 2024
Baylor University; Waco, TX
Held supplemental instruction lectures to assist students in their introductory statistics coursework. Assisted professors with grading assessments.
Fall 2023 & Spring 2024 - STA 1380: Introduction to Statistics

Instructor of Record (Graduate Student) Aug 2022 - May 2023
Texas Tech University; Lubbock, TX
Fully responsible for sections of business mathematics and linear algebra, developing lectures, assessments, and holding regular office hours for 60-130 students per term.
Fall 2022 - MATH 1330 116: Introductory Mathematical Analysis 1
Spring 2023 - MATH 2360 012: Linear Algebra & MATH 2360 021: Linear Algebra

Graduate Assistant - Teaching Assistant & Tutor Aug 2021 - Aug 2022
Texas Tech University; Lubbock, TX
Graded homework assignments and assessments for calculus and abstract algebra courses. Tutored students in a variety of topics such as statistics, algebra, and calculus.
Fall 2021 - MATH 2450: Calculus III with Applications
Spring 2022 - MATH 3360: Foundations of Algebra 1
Summer 2022 - Department of Mathematics and Statistics Tutor

Department of Mathematics Tutor Jan 2019 - May 2021
University of Mary Hardin-Baylor; Belton, TX
Assisted students in understanding a variety of topics such as statistics, algebra, and calculus.

Research Experience

Research Assistant - Baylor Water Security Collaborative Jan 2026 - Present
Baylor University; Waco, TX
Part of an interdisciplinary group of researchers working on water-related problems. With my advisor, inventory and characterize public water data resources across Texas, documenting coverage, access methods, and citation practices to identify statewide data gaps. In the early stages of framing this inventory as a network analysis, with the organizations that collect, aggregate, and use water data as nodes and the data flows between them as connections. Programmatically retrieve and analyze hydrologic time series from public sources.

Master's Report April 2023
Texas Tech University; Lubbock, TX
Advisor: Dr. Lourdes Juan
Neural Networks in Machine Learning: Comparing Activation Functions

Undergraduate Thesis April 2021
University of Mary Hardin-Baylor; Belton, TX
Advisor: Dr. Peter Chen
Bootstrap-based Error Rate Estimators for the Linear Discriminant function
- Awarded "Best Presentation in Applied Mathematics, Probability and Statistics" at the 2021 Annual Meeting of the Texas Section of the MAA.

Presentations & Conferences

Taylor, N. C., & Hering, A. S. (2026, June). Predicting in-lake turbidity using satellite data: A comparison of spatial models [Presented at the UCOWR/NIWR Annual Water Resources Conference; San Antonio, TX].

Taylor, N. C., & Chen, P. (2021, April). Bootstrap-based error rate estimators for the linear discriminant function [Presented at the Annual Meeting of the Texas Section of the MAA; Virtual via Zoom].

Technical Skills

- **Statistical Modeling:** spatial & spatiotemporal models (kriging, Bayesian INLA), regression, cross-validation design, simulation studies, bootstrap methods, multivariate process monitoring, neural networks, functional data analysis
- **Programming & Tools:** R, Python, MATLAB, SQL, JMP, Git/GitHub, Google Earth Engine, LaTeX
- **Data:** satellite imagery (Sentinel-2), geospatial data, hydrologic time series, high-volume sensor data

Organizations & Honors

- Member of the American Statistical Association (ASA)
- Baylor's Baptist College & University (BCU) Scholars
- The Honor Society of Phi Kappa Phi
- Delta Psi Theta Math Club - Treasurer (Spring 2020) & Secretary (Fall 2020)
- Gamma Beta Phi Society
- Kappa Mu Epsilon National Mathematics Honor Society
- Alpha Chi National College Honor Society

Athletics

Texas Tech University Powerlifting Club — Jan 2022 - Jan 2023

University of Mary Hardin-Baylor Football Team — Aug 2018 - Feb 2021
2018 NCAA Division III National Champions